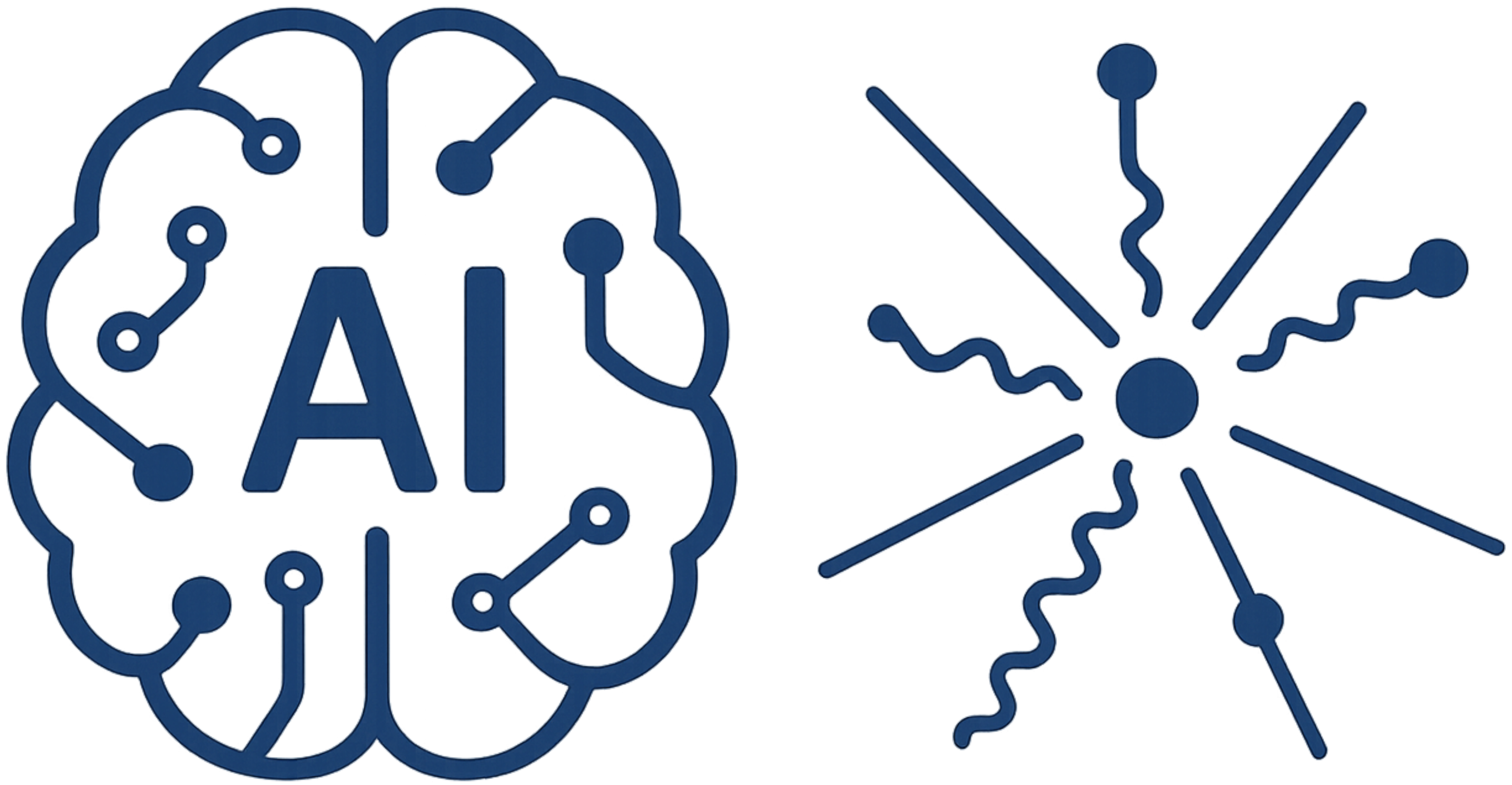
Introduction to

AI-Driven HEP

S. A. Fard - School of Physics (IPM)



AI-Driven HEP 7

Session 7

Statistical Learning II

Reference

[1] James, Gareth, et al. "Statistical learning." An introduction to statistical learning: With applications in Python. (2023)

[2] Hastie, Trevor. "The elements of statistical learning: data mining, inference, and prediction." (2009)

[3] Bishop, Christopher M., Pattern recognition and machine learning. springer, (2006)

[4] Chollet, Francois. Deep learning with Python. (2021)

[5] Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. (2014)

[6] Tingjun Yang, A Brief Description of Data Products, ProtoDUNE Analysis Workshop, Jan 27, 2019, CERN

[7] Gabriela Vitti Stenico, Development of the computing framework for monitoring the quality of ProtoDUNE off-line data, DUNE collaboration meeting (2025)

What does an Input look like

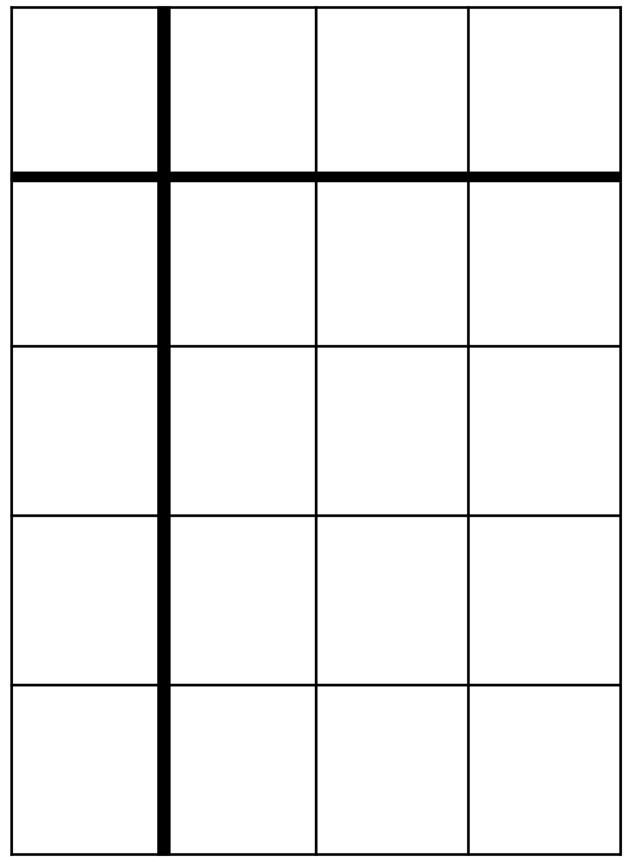
An array with shape (n, m, l, k)

(n, 1, 1, k)

(n, 1, l, k)

(n, m, l, k)

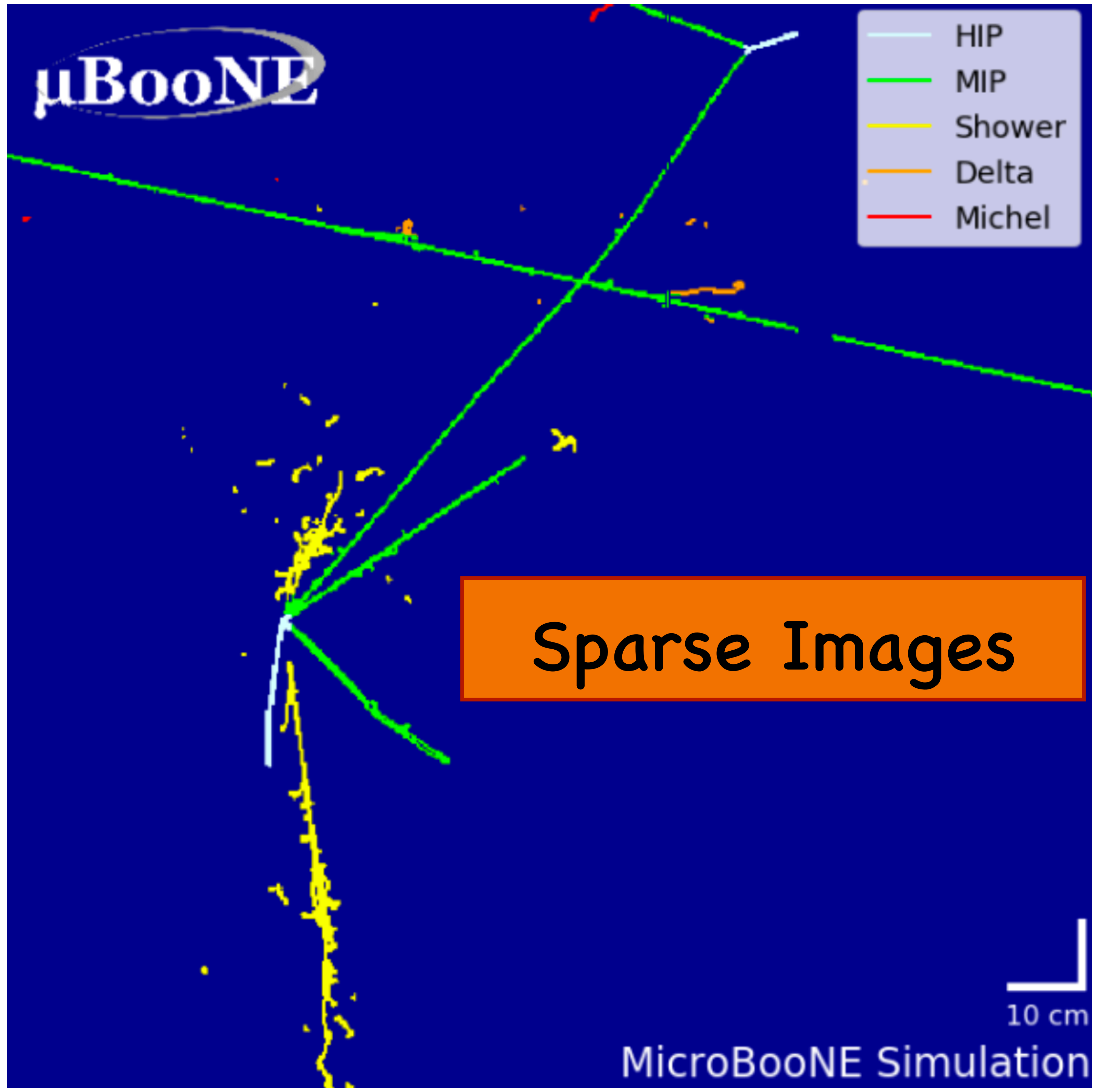
n k



What does an Input look like

The organized (m, l, k) data format

no longer works



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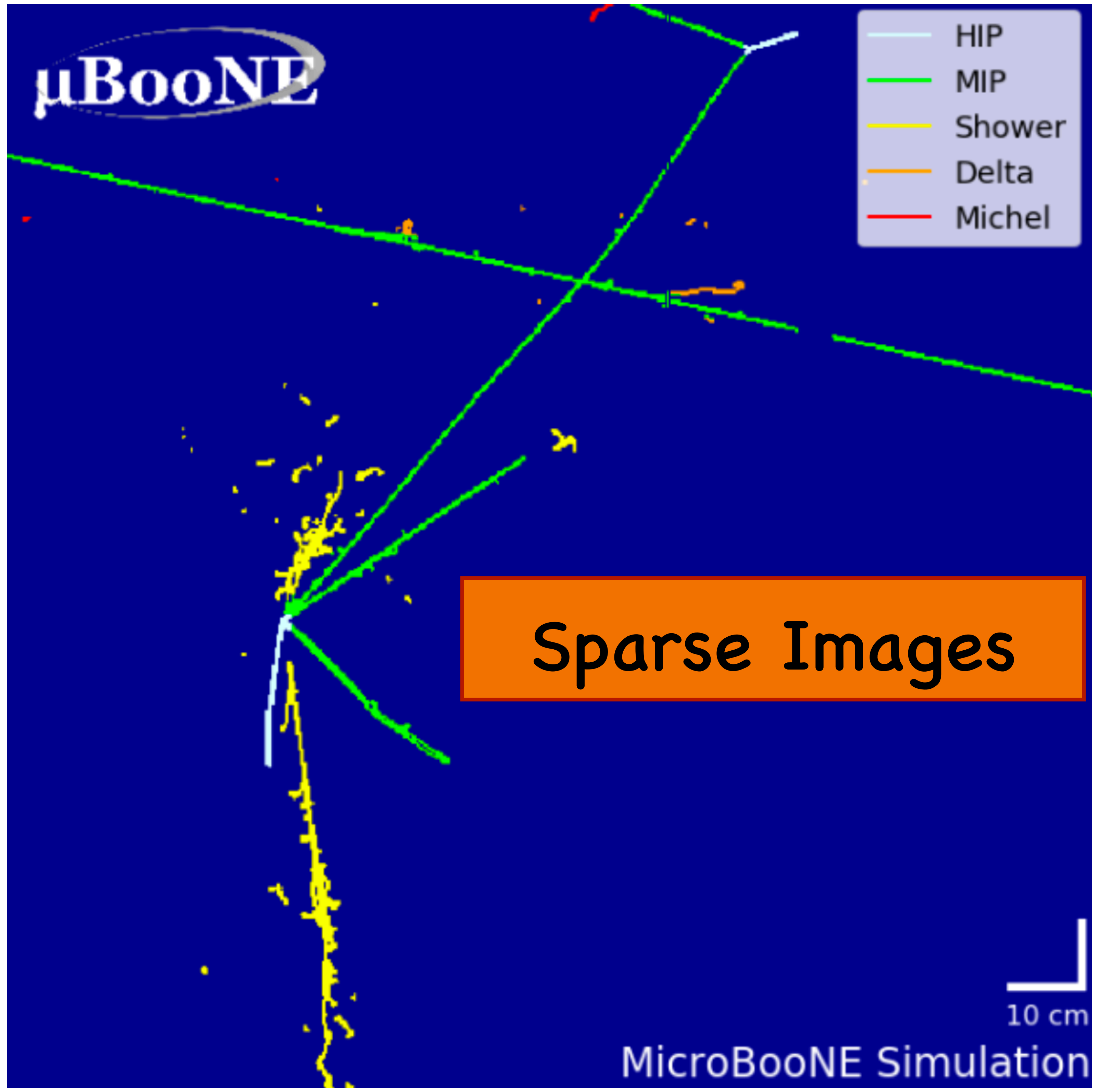

Sparse Images

2.1 What Is Statistical Learning?

In order to motivate our study of statistical learning, we begin with a simple example. Suppose that we are statistical consultants hired by a client to investigate the association between advertising and sales of a particular product. The `Advertising` data set consists of the `sales` of that product in 200 different markets, along with advertising budgets for the product in each of those markets for three different media: `TV`, `radio`, and `newspaper`. The data are displayed in Figure 2.1. It is not possible for our client to directly increase sales of the product. On the other hand, they can control the advertising expenditure in each of the three media. Therefore, if we determine that there is an association between advertising and sales, then we can instruct our client to adjust advertising budgets, thereby indirectly increasing sales. In other words, our goal is to develop an accurate model that can be used to predict sales on the basis of the three media budgets.

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More generally, suppose that we observe a quantitative response Y and p different predictors, $X_1, X_2, \ldots, X_p$. We assume that there is some relationship between Y and $X = (X_1, X_2, \ldots, X_p)$, which can be written in the very general form

$$Y = f(X) + \epsilon. \quad (2.1)$$

Set of words sequence

AI-Driven HEP 7

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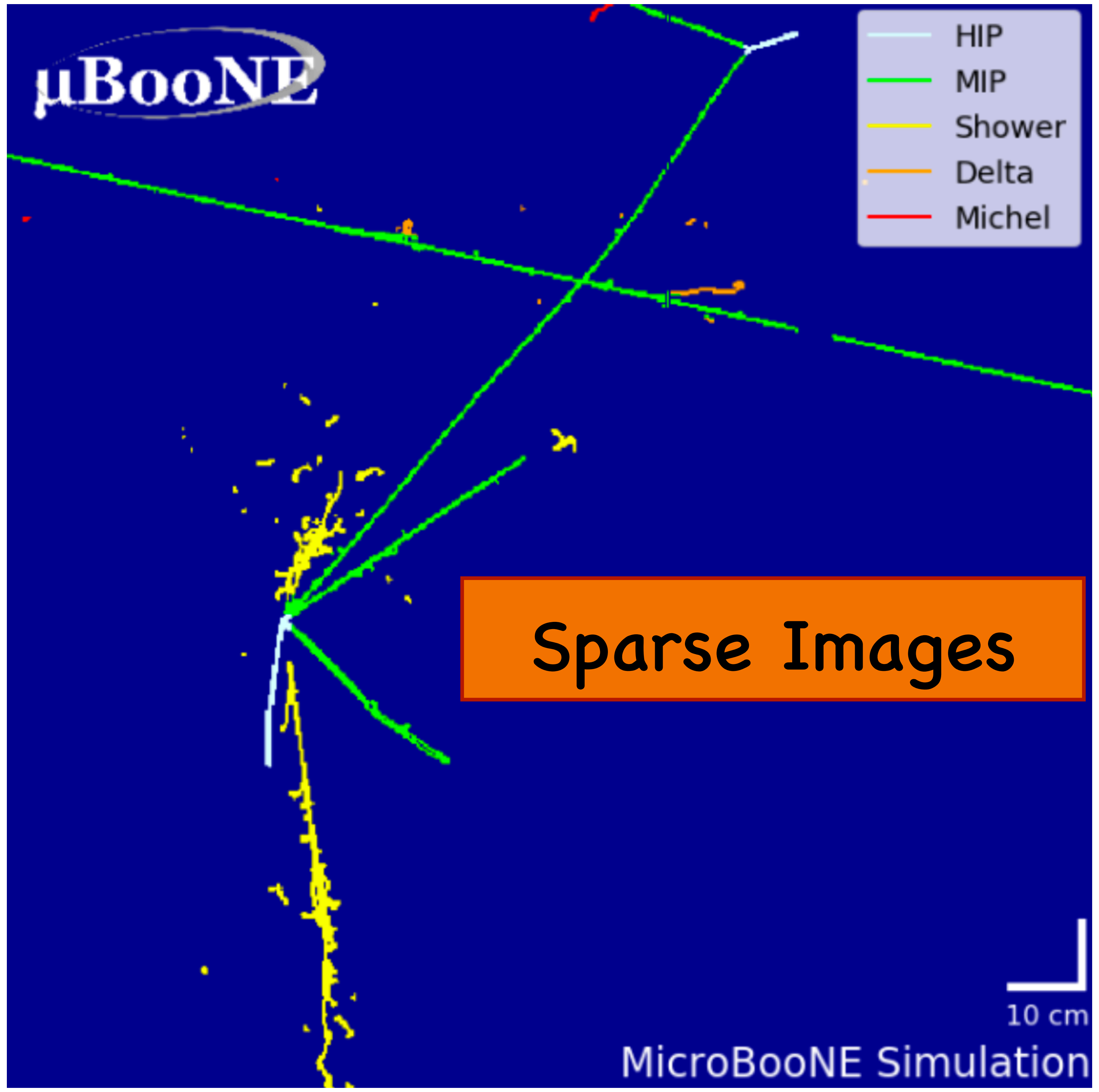

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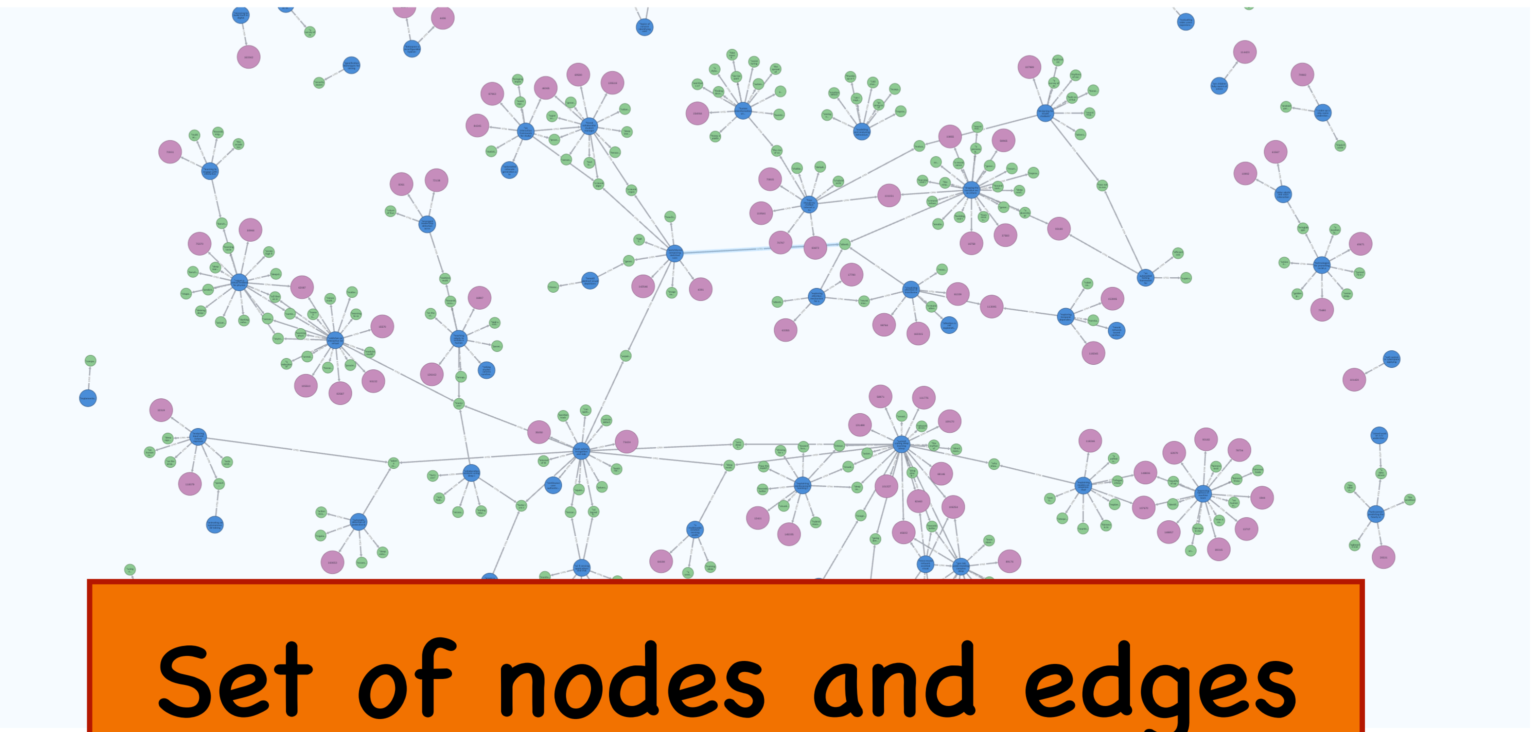

Set of nodes and edges

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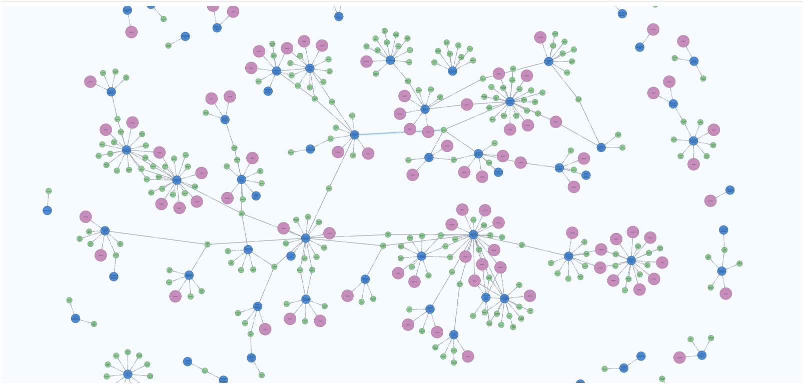

Graph format

	c1	c2	c3
r1	46	73	12
r2	89	19	35
r3	54	62	3

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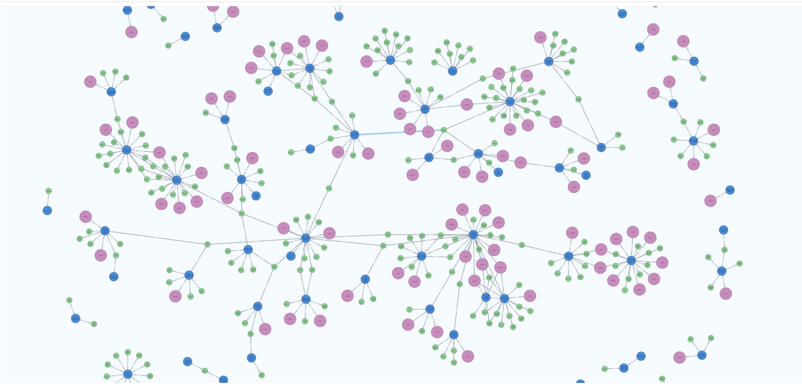

Graph format

	c1	c2	c3
r1	46	73	12
r2	89	19	35
r3	54	62	3

$$\begin{array}{cc} \begin{array}{c} 11 \\ 12 \\ 13 \\ 21 \\ 22 \\ 23 \\ 31 \\ 32 \\ 33 \end{array} & X = \begin{bmatrix} 46 \\ 73 \\ 12 \\ 89 \\ 19 \\ 35 \\ 54 \\ 62 \\ 3 \end{bmatrix} \end{array}$$

AI-Driven HEP 7

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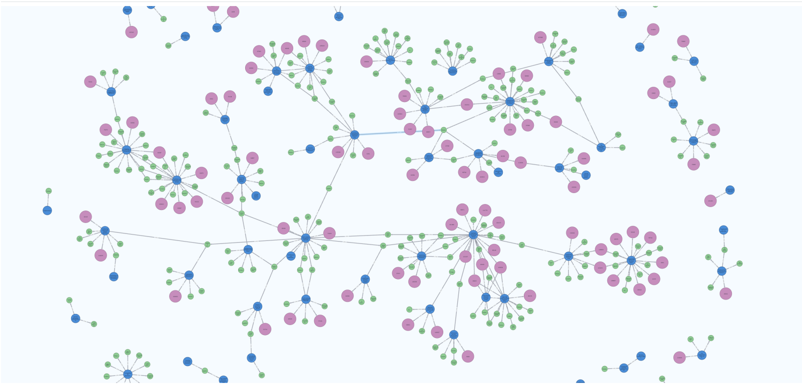

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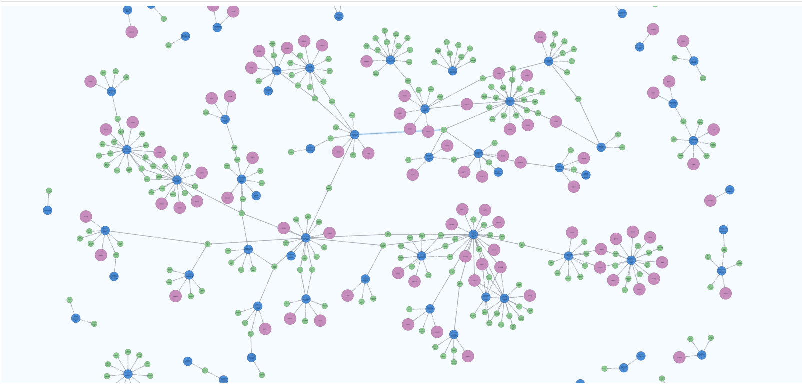
$$
\begin{array}{c} 11 \\ 12 \\ 13 \\ 21 \\ 22 \\ 23 \\ 31 \\ 32 \\ 33 \end{array}
\quad
X = \begin{bmatrix} 46 \\ 73 \\ 12 \\ 89 \\ 19 \\ 35 \\ 54 \\ 62 \\ 3 \end{bmatrix}
\qquad
\begin{array}{c}
\begin{array}{ccccccccc} 11 & 12 & 13 & 21 & 22 & 23 & 31 & 32 & 33 \end{array} \\
A = \begin{bmatrix}
1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\
1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\
0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\
0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1
\end{bmatrix}
\end{array}
$$

What does an Input look like



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r1	46	73	12
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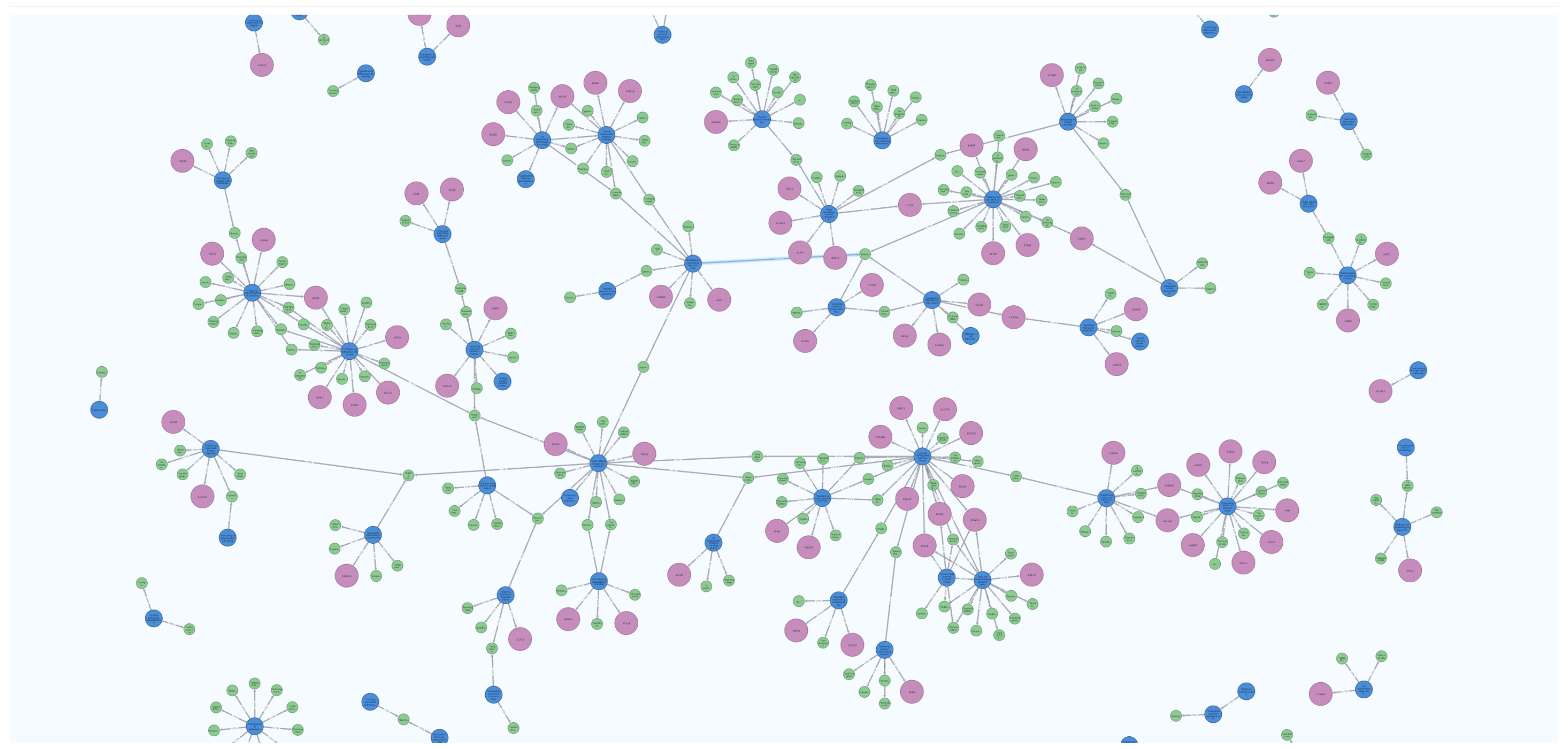
$$\begin{array}{c} 11 \\ 12 \\ 13 \\ 21 \\ 22 \\ 23 \\ 31 \\ 32 \\ 33 \end{array} \quad X = \begin{bmatrix} 46 \\ 73 \\ 12 \\ 89 \\ 19 \\ 35 \\ 54 \\ 62 \\ 3 \end{bmatrix} \qquad A = \begin{array}{c} \begin{array}{ccccccccc} 11 & 12 & 13 & 21 & 22 & 23 & 31 & 32 & 33 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

$$E = \begin{bmatrix} (11,11) : w_1 \\ (11,12) : w_2 \\ \vdots \\ (33,33) : w_{33} \end{bmatrix}$$

What does an Input look like

Graph format



Relax the condition

$$
\begin{array}{c}
11 \\ 12 \\ 13 \\ 21 \\ 22 \\ 23 \\ 31 \\ 32 \\ 33
\end{array}
\quad
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\qquad
\begin{array}{c}
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1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
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1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\
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\end{bmatrix}
\end{array}
$$

What does an Input look like

Text format

From natural languages to machine language

2.1 What Is Statistical Learning?

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AI-Driven HEP 7

What does an Input look like

Text format

From natural languages to machine language

How much the sequence of the words affect our understanding ?

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How the similarities of the words are crucial ?

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How much the sequence of the words affect our understanding ?

How the similarities of the words are crucial ?

How the context of the words play roles ?

AI-Driven HEP 7

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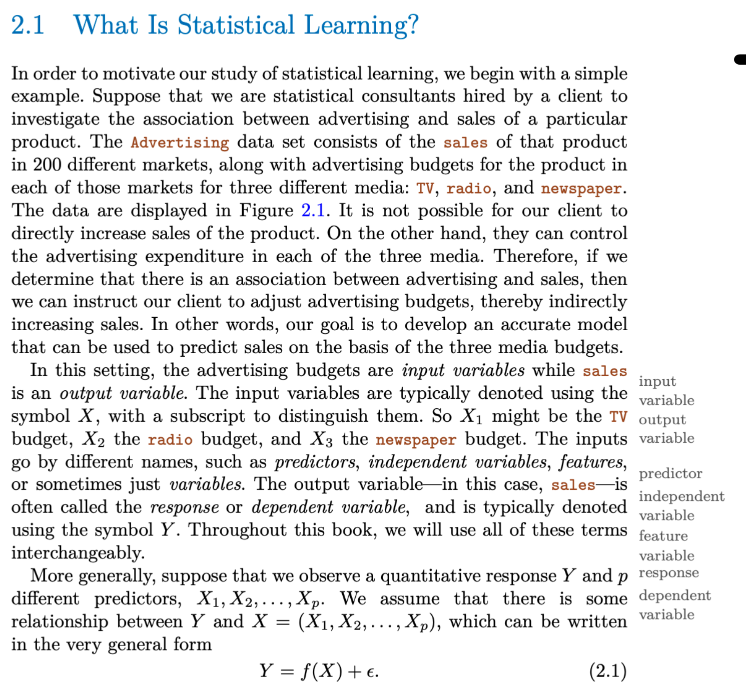

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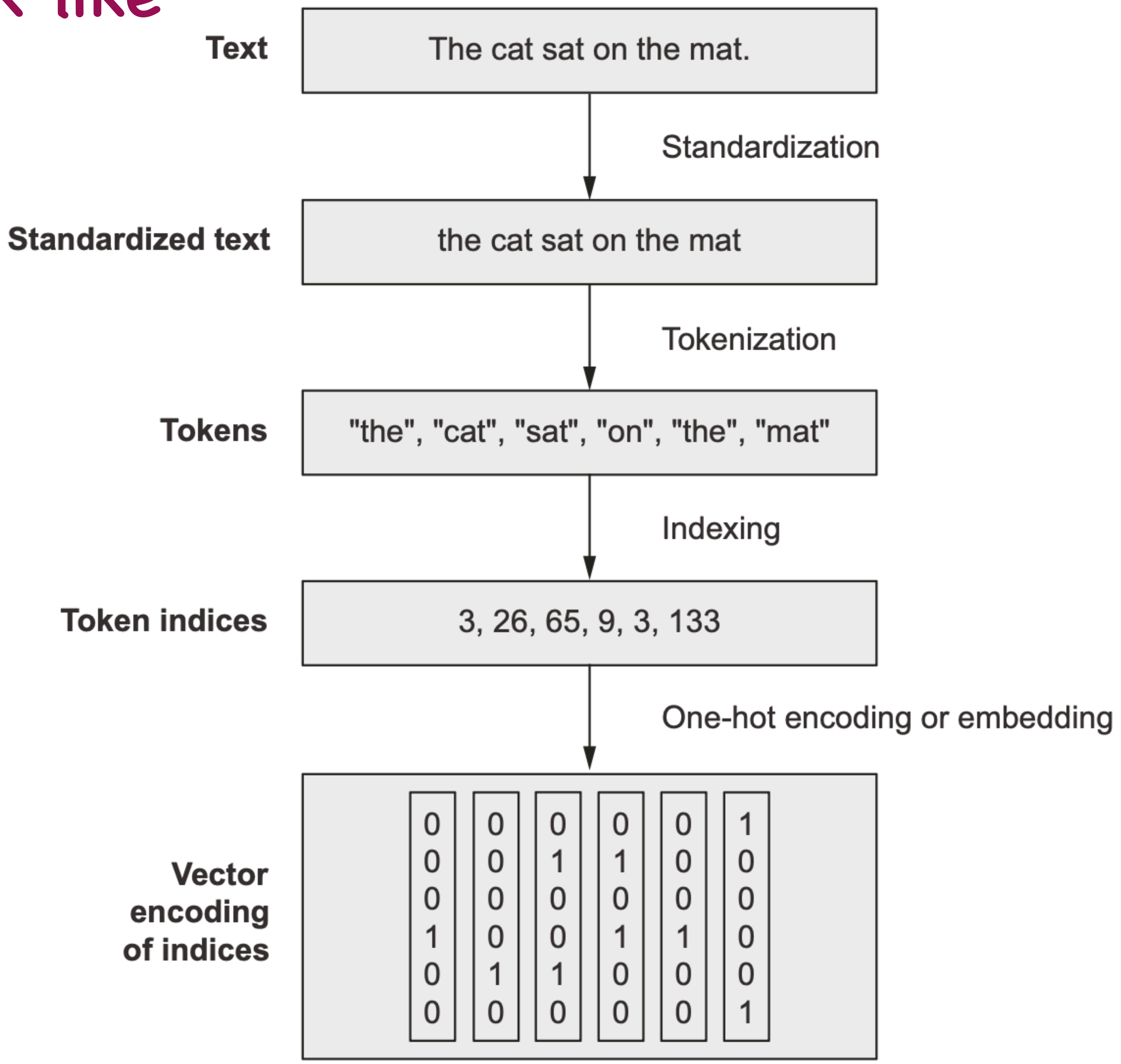

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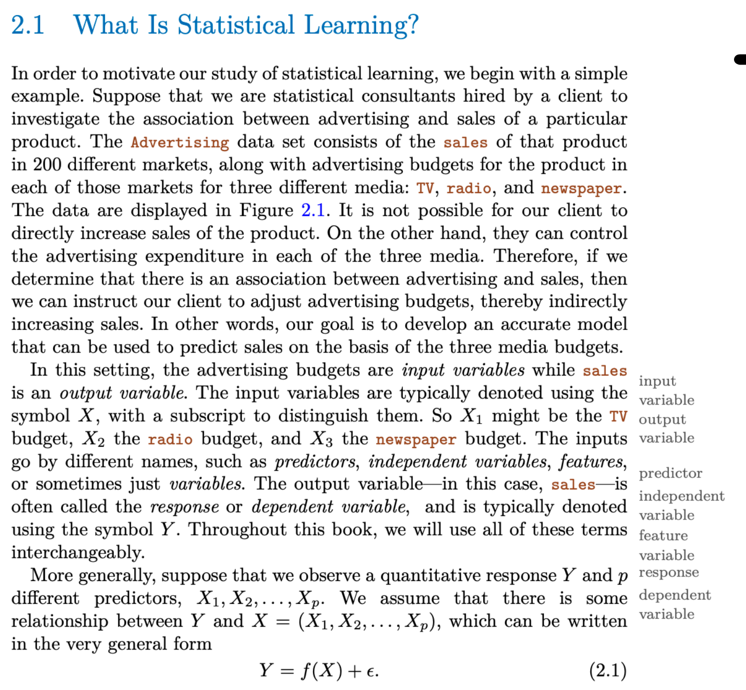

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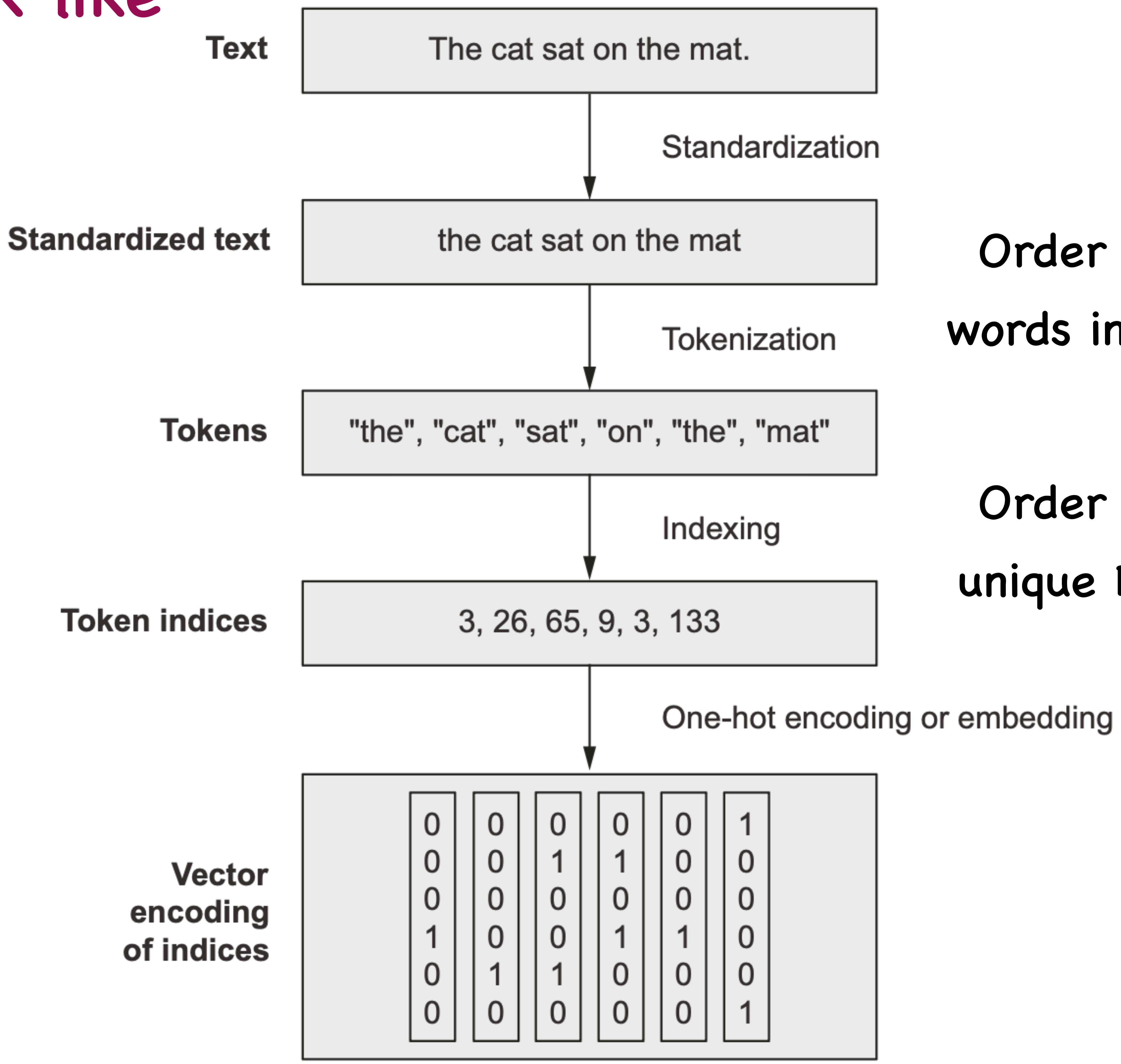
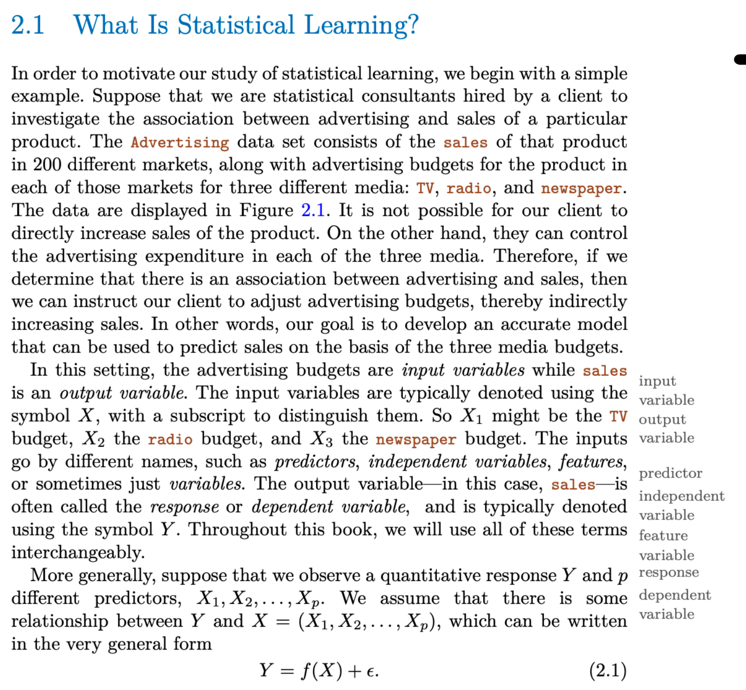


Order of total number of words in entire web $\sim 10^{15}$

Order of total number of unique English word $\sim 10^5$

What does an Input look like



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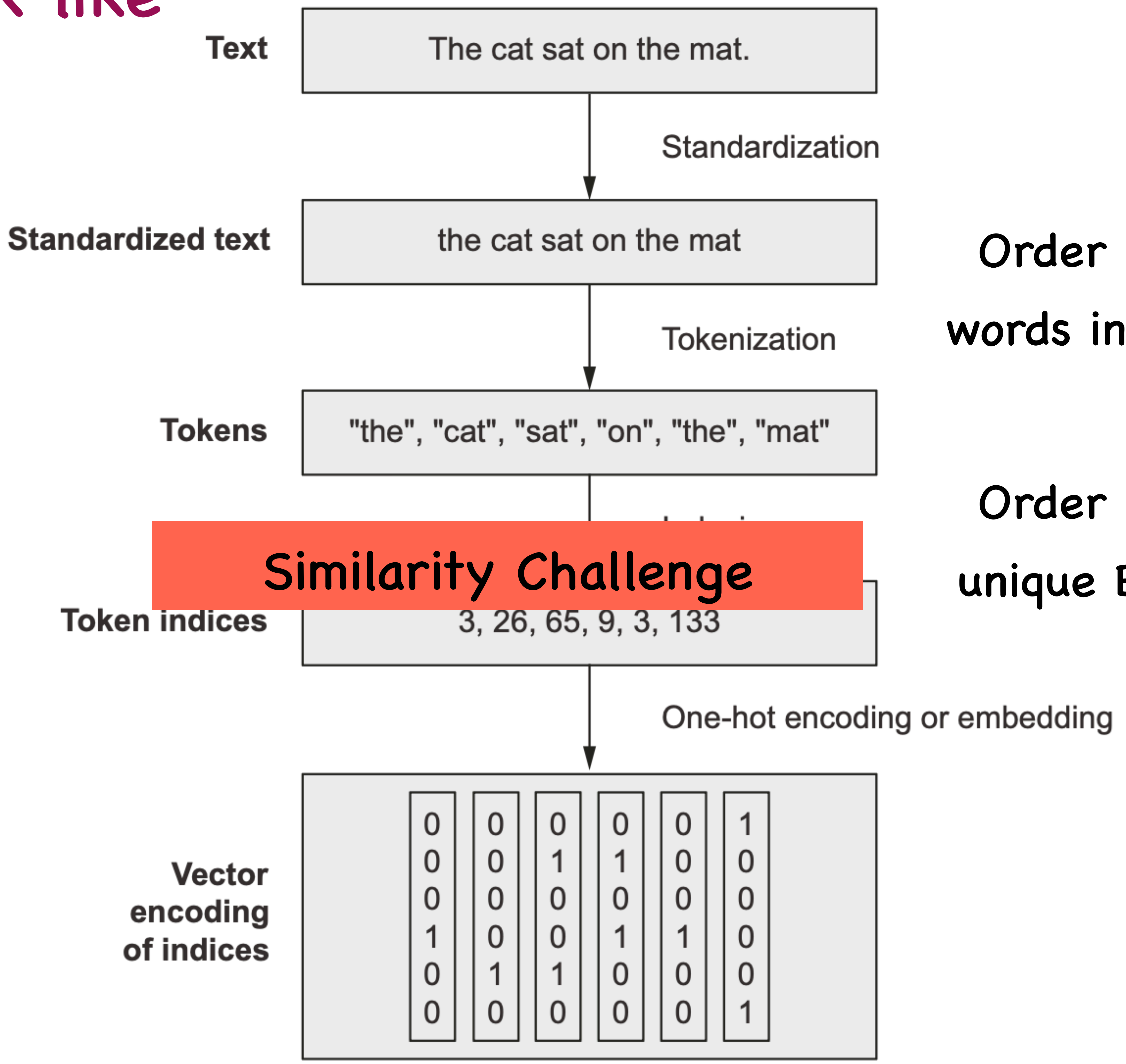


Similarity Challenge

Order of total number of words in entire web $\sim 10^{15}$

Order of total number of unique English word $\sim 10^{5}$

What does an Input look like

An array with shape (n, m, l, k)

Domain of the data

AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain of the data

Incompleteness and generalisation

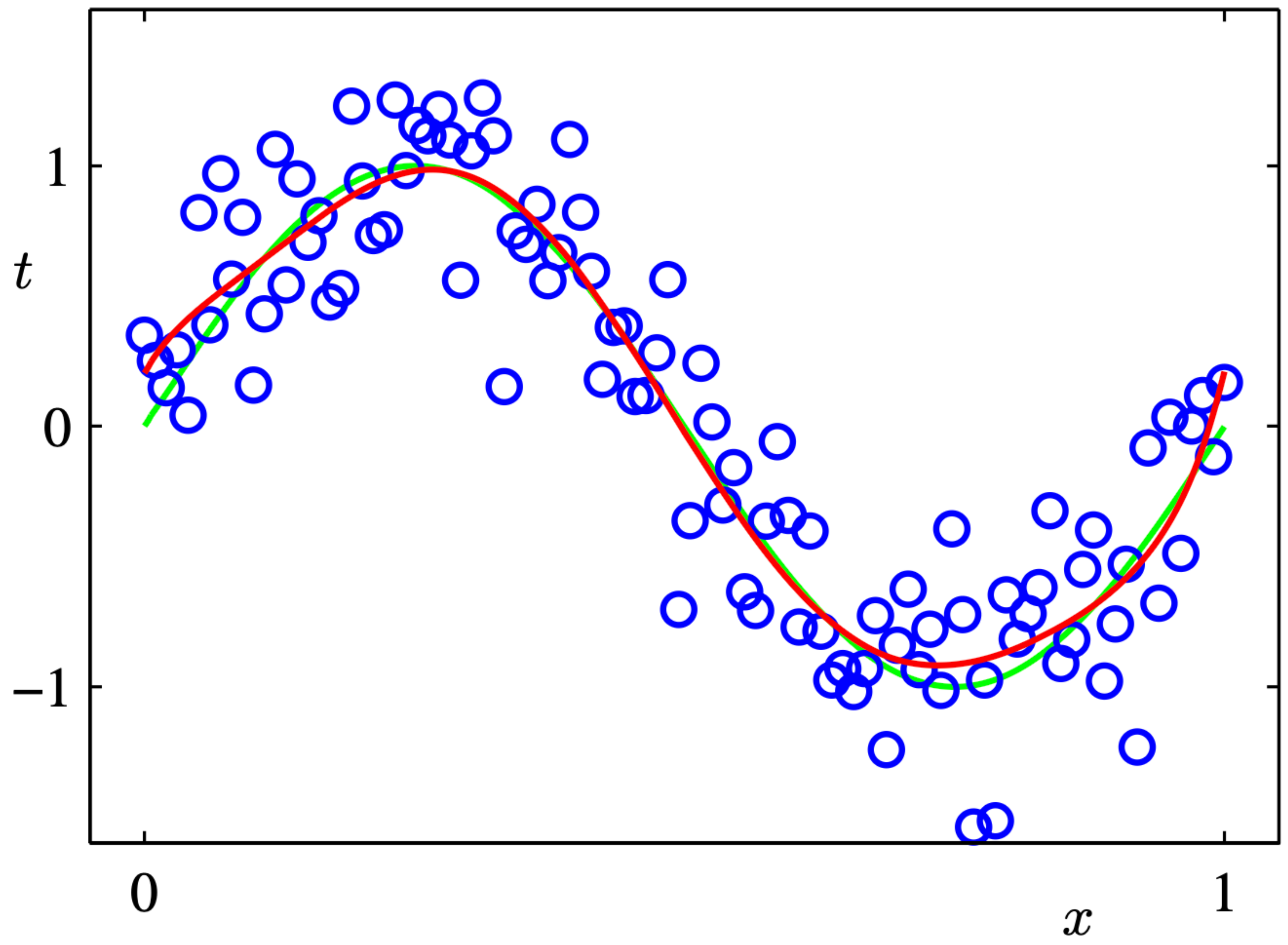
AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain of the data

Incompleteness and generalisation

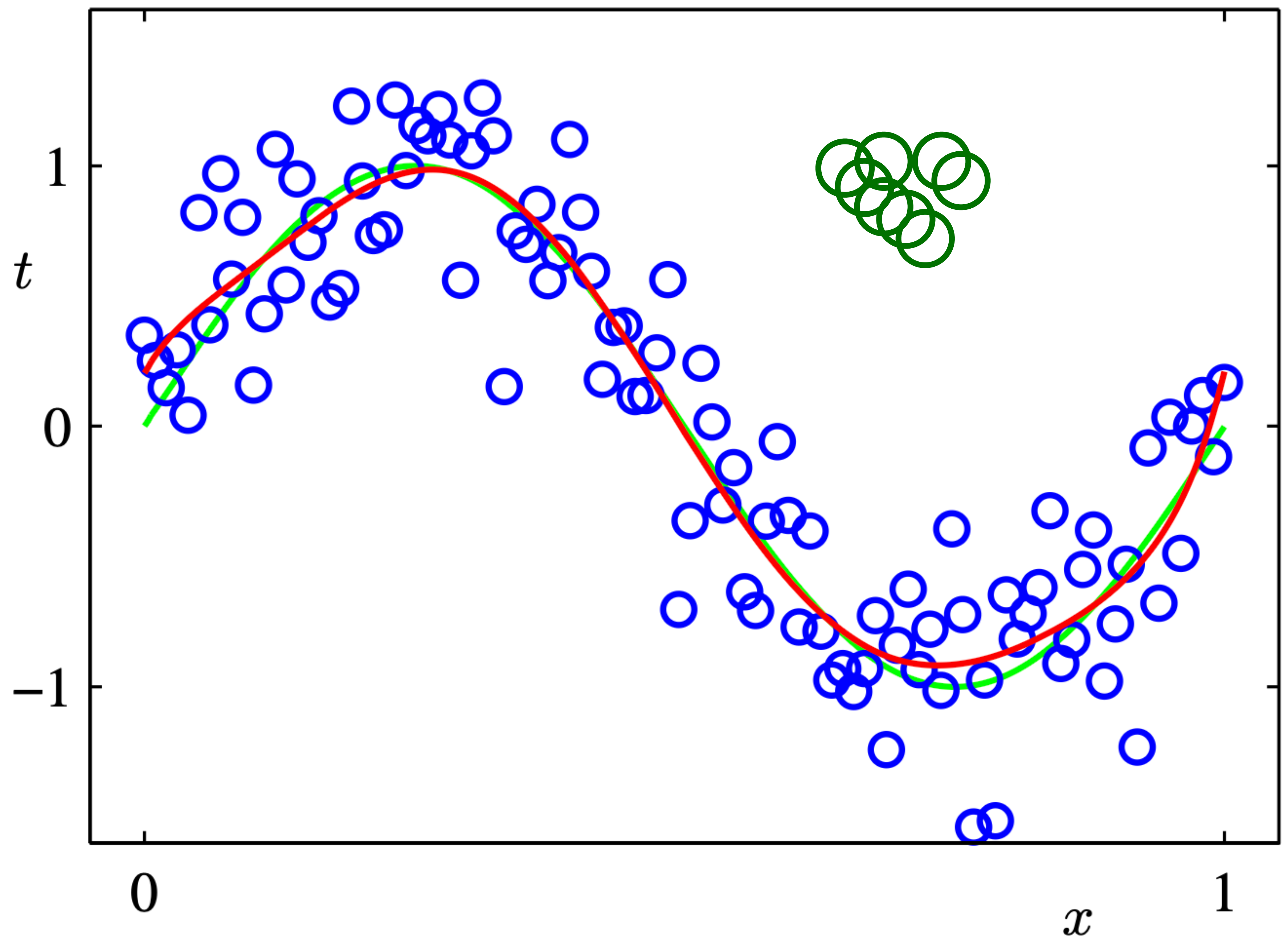


What does an Input look like

An array with shape (n, m, l, k)

Domain of the data

Incompleteness and generalisation

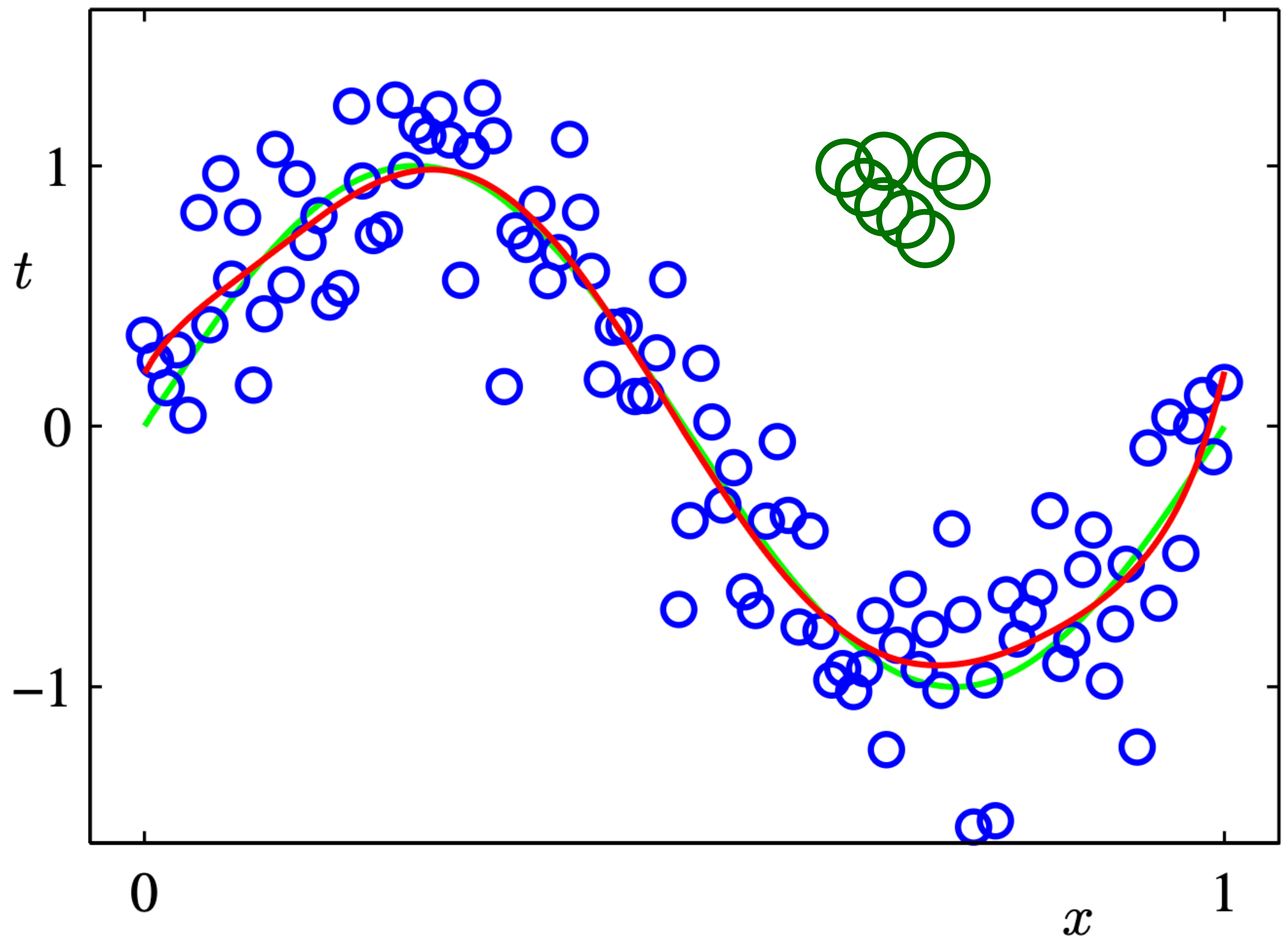


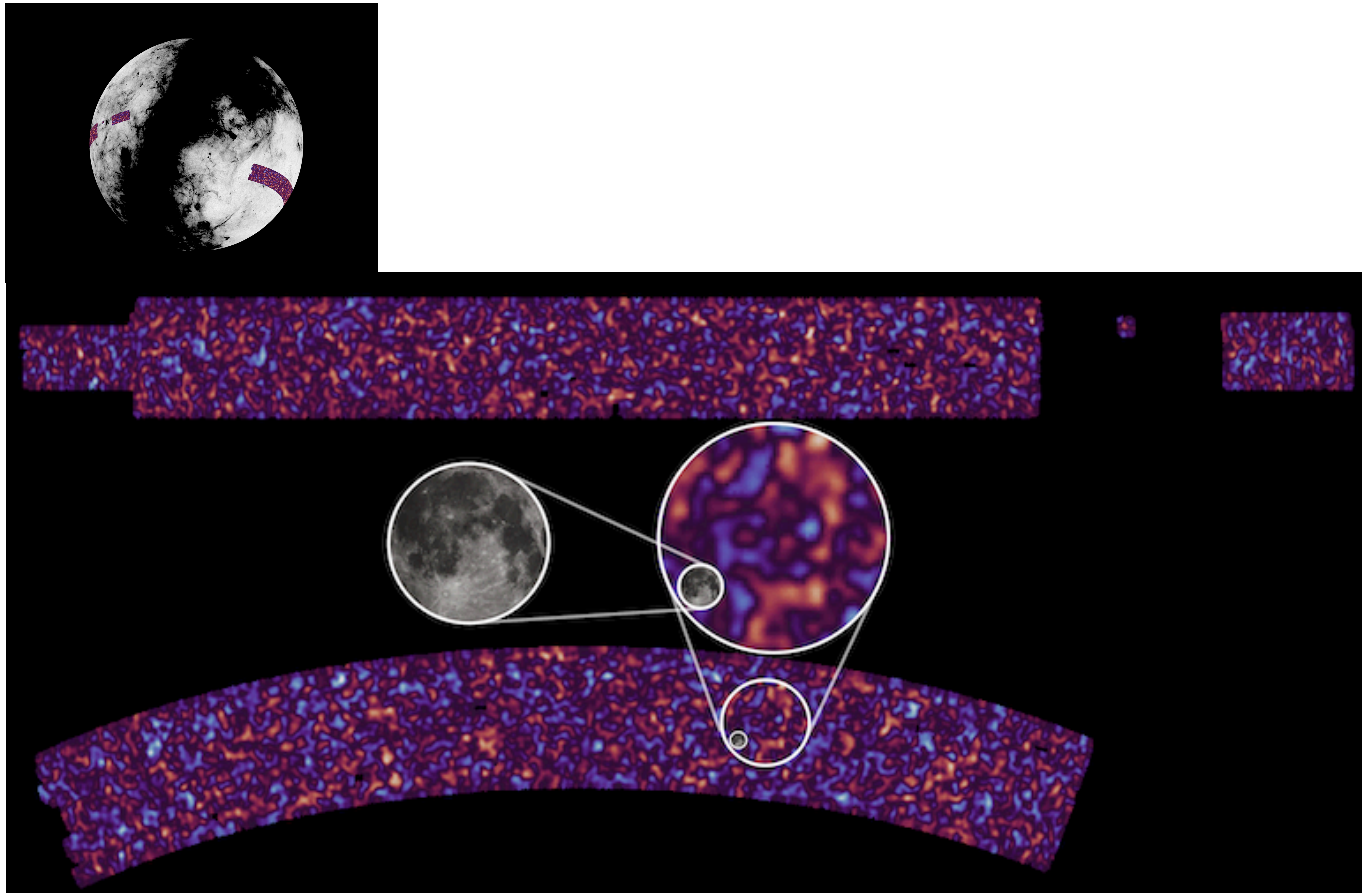
What does an Input look like

An array with shape (n, m, l, k)

Domain of the data

Incompleteness and generalisation





https://kids.strw.leidenuniv.nl/

What does an Input look like

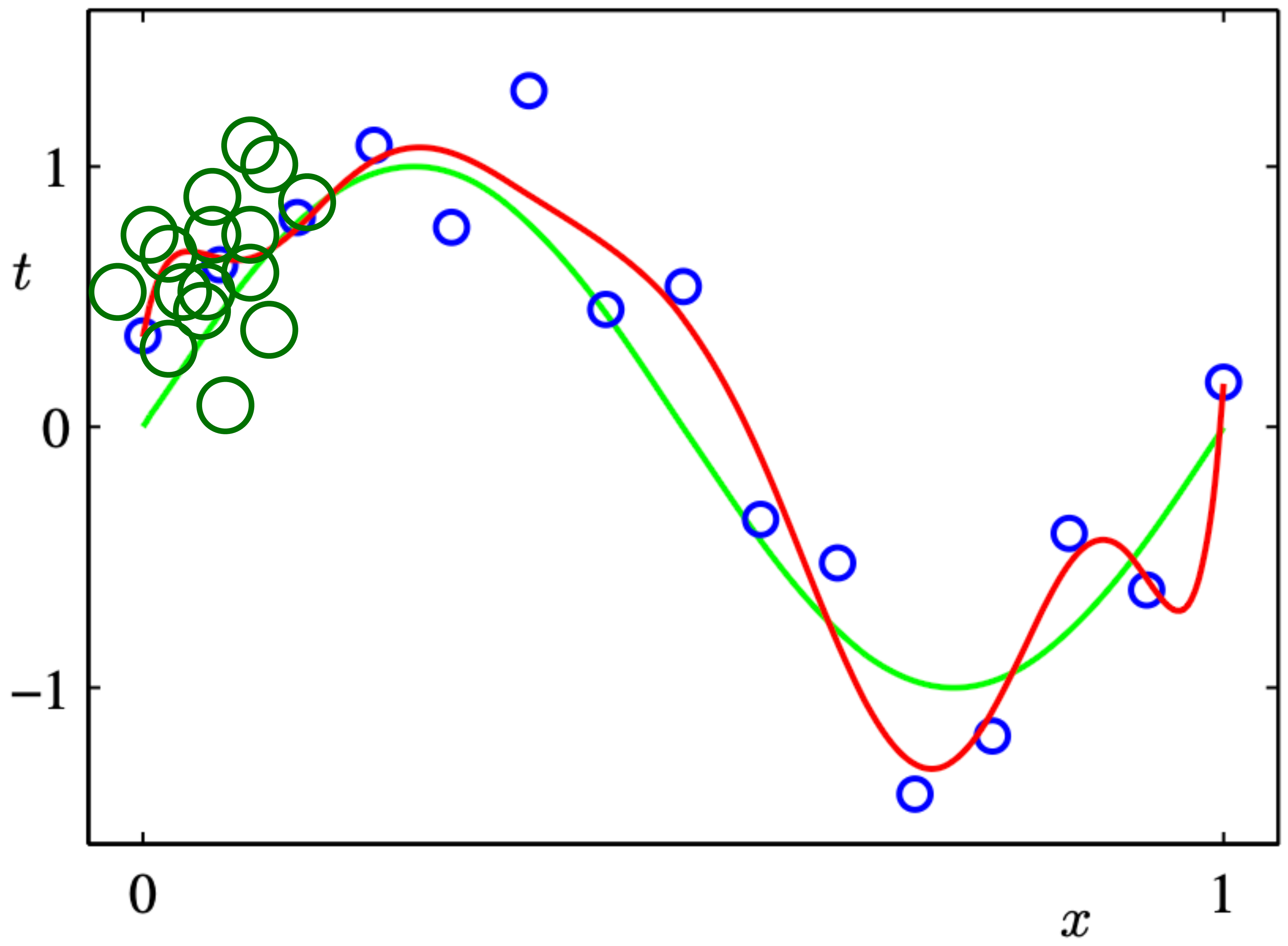
An array with shape (n, m, l, k)

Domain Distribution

What does an Input look like

An array with shape (n, m, l, k)

Domain Distribution

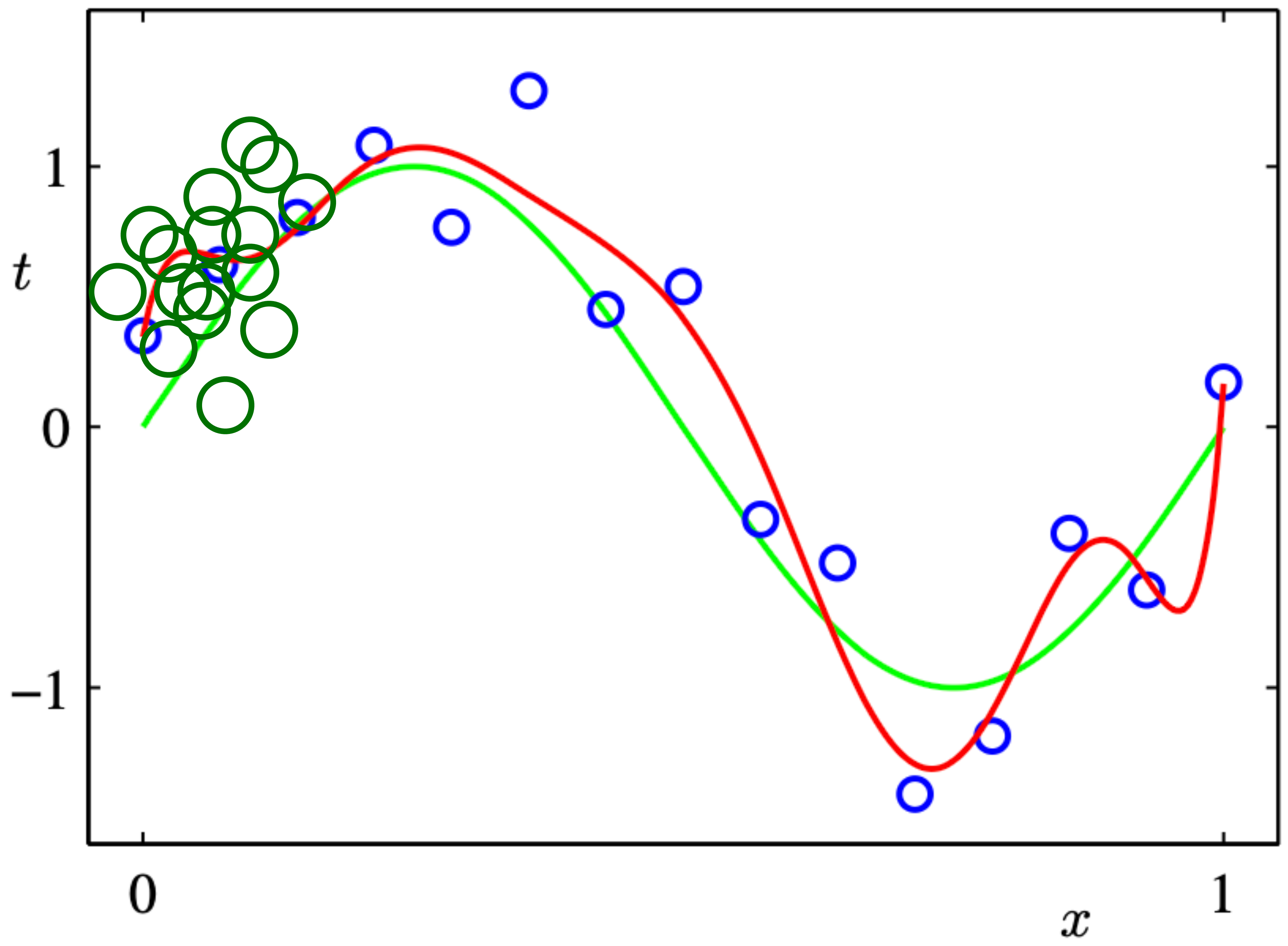


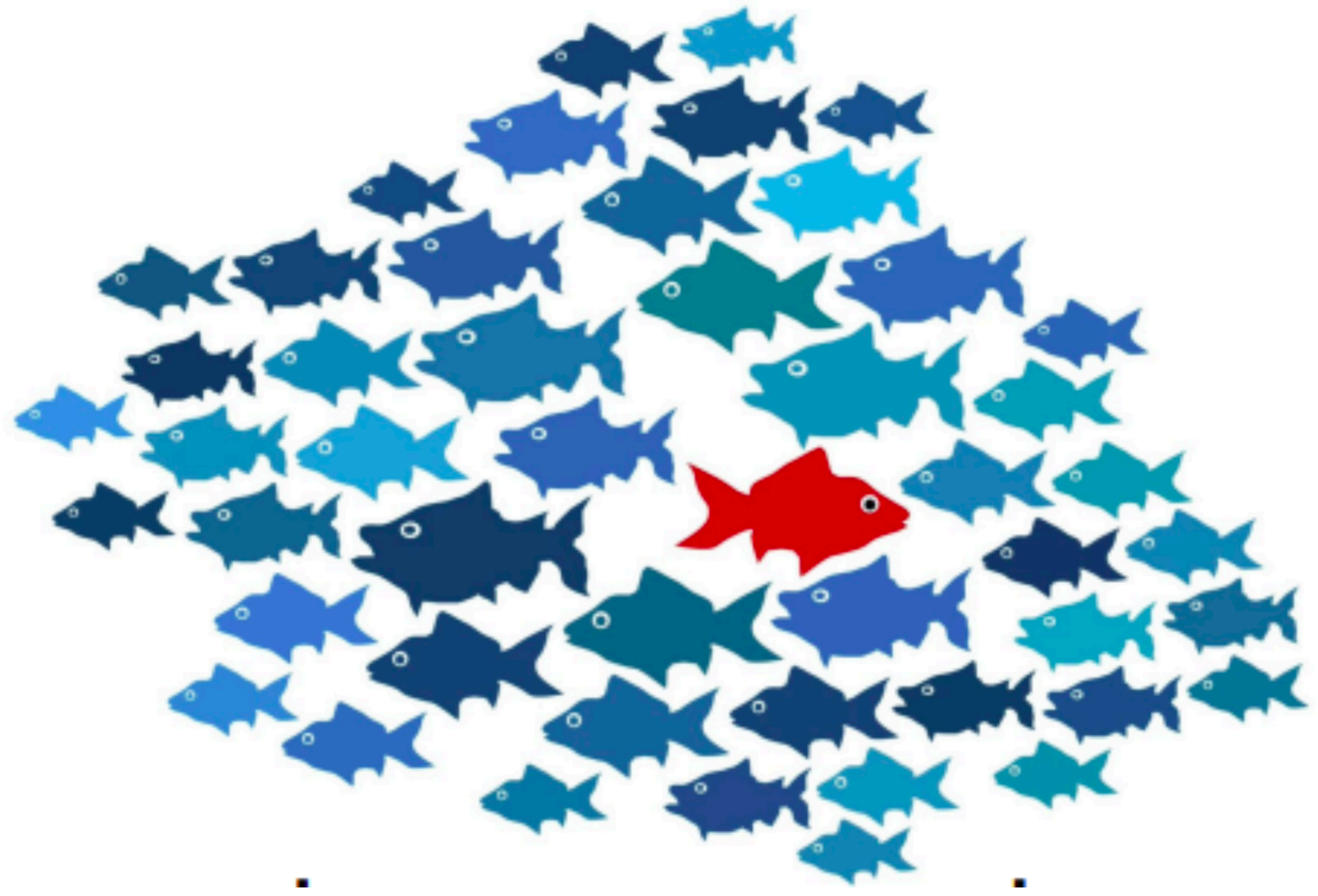
AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain Distribution





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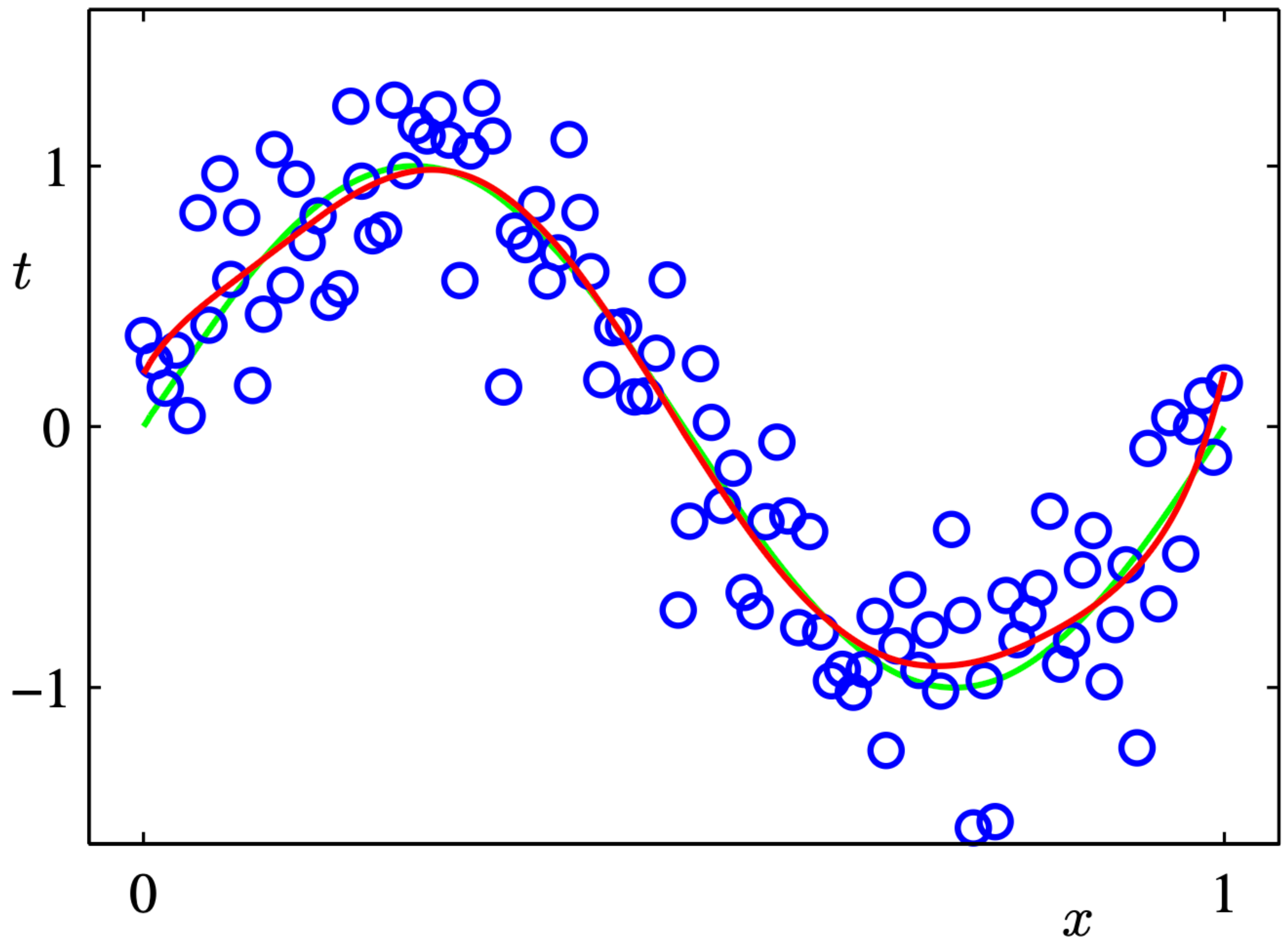
What does an Input look like

An array with shape (n, m, l, k)

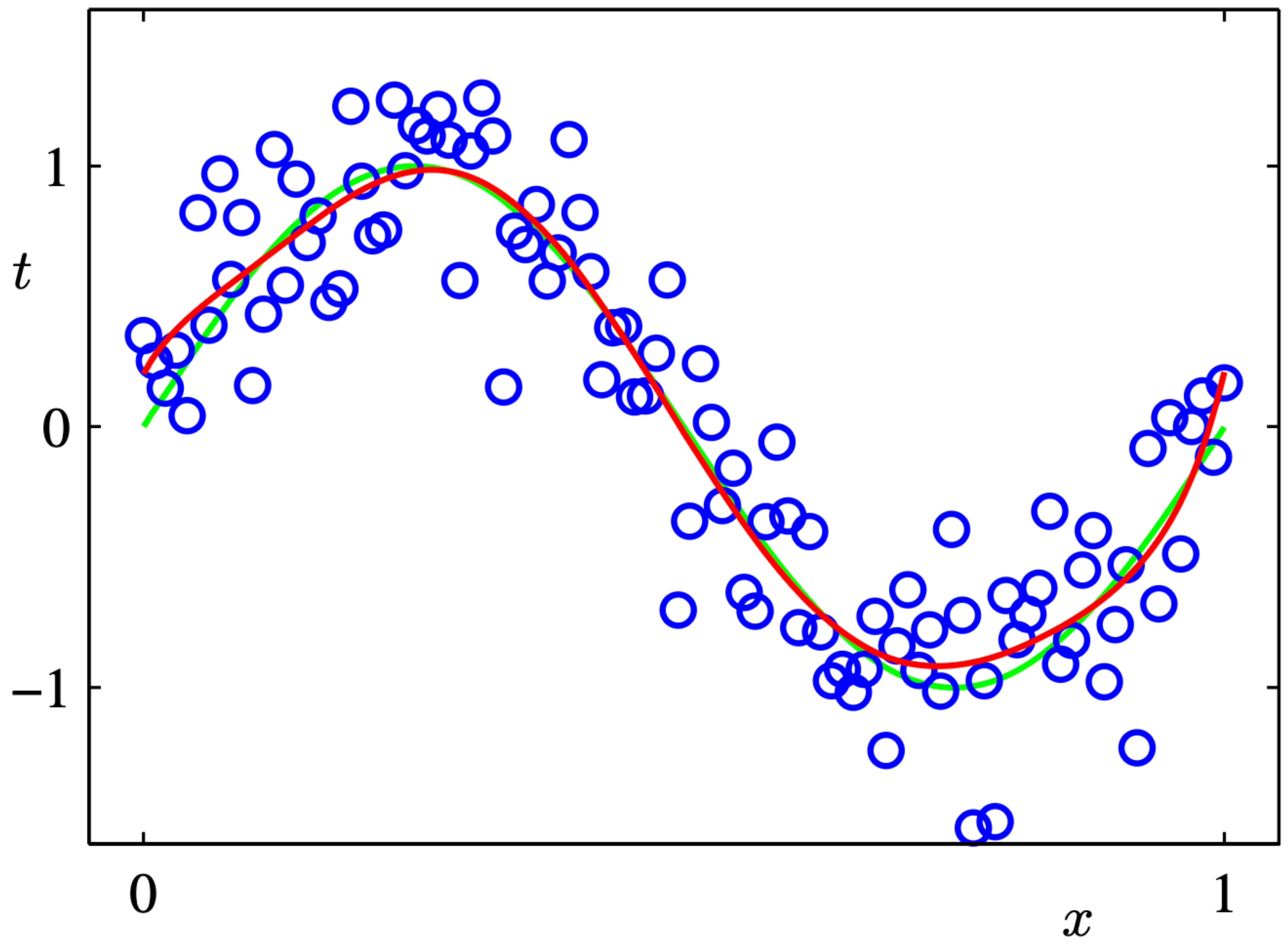
Domain Uncertainty



What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty I.I.D



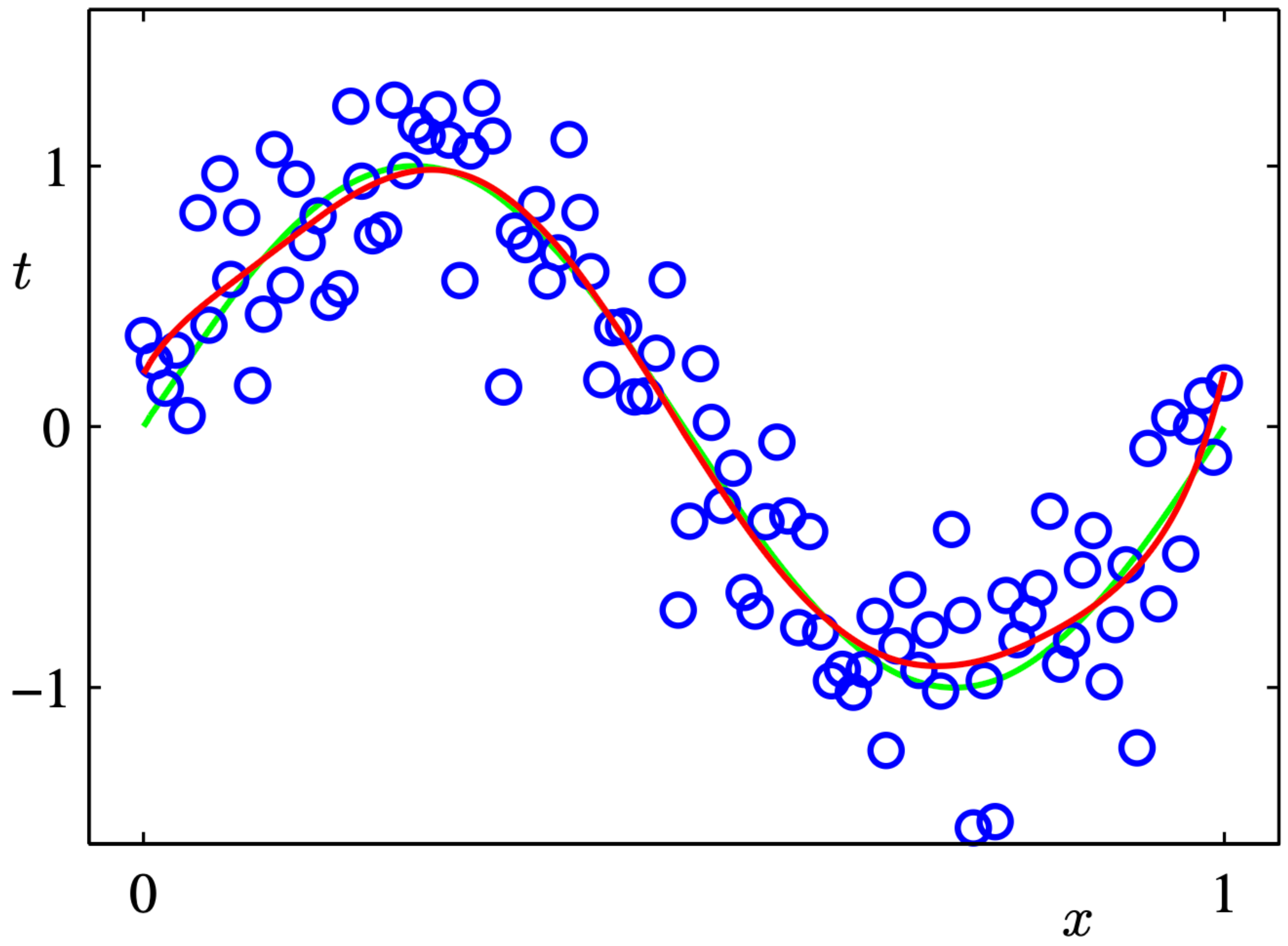
AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty **I.I.D**

$$\mathcal{N}\left(x|\mu,\sigma^2\right) = \frac{1}{(2\pi\sigma^2)^{1/2}}\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$



AI-Driven HEP 7

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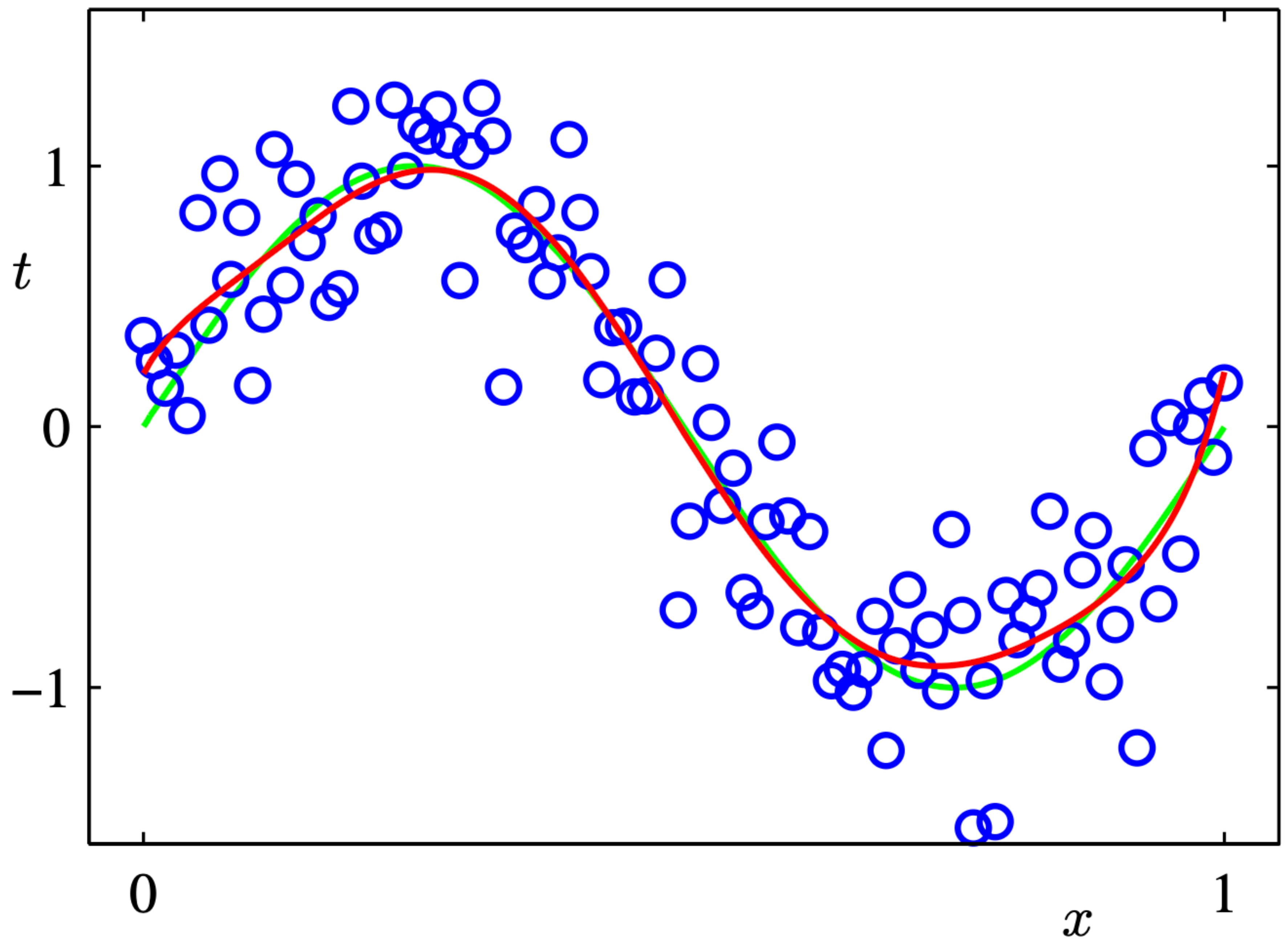


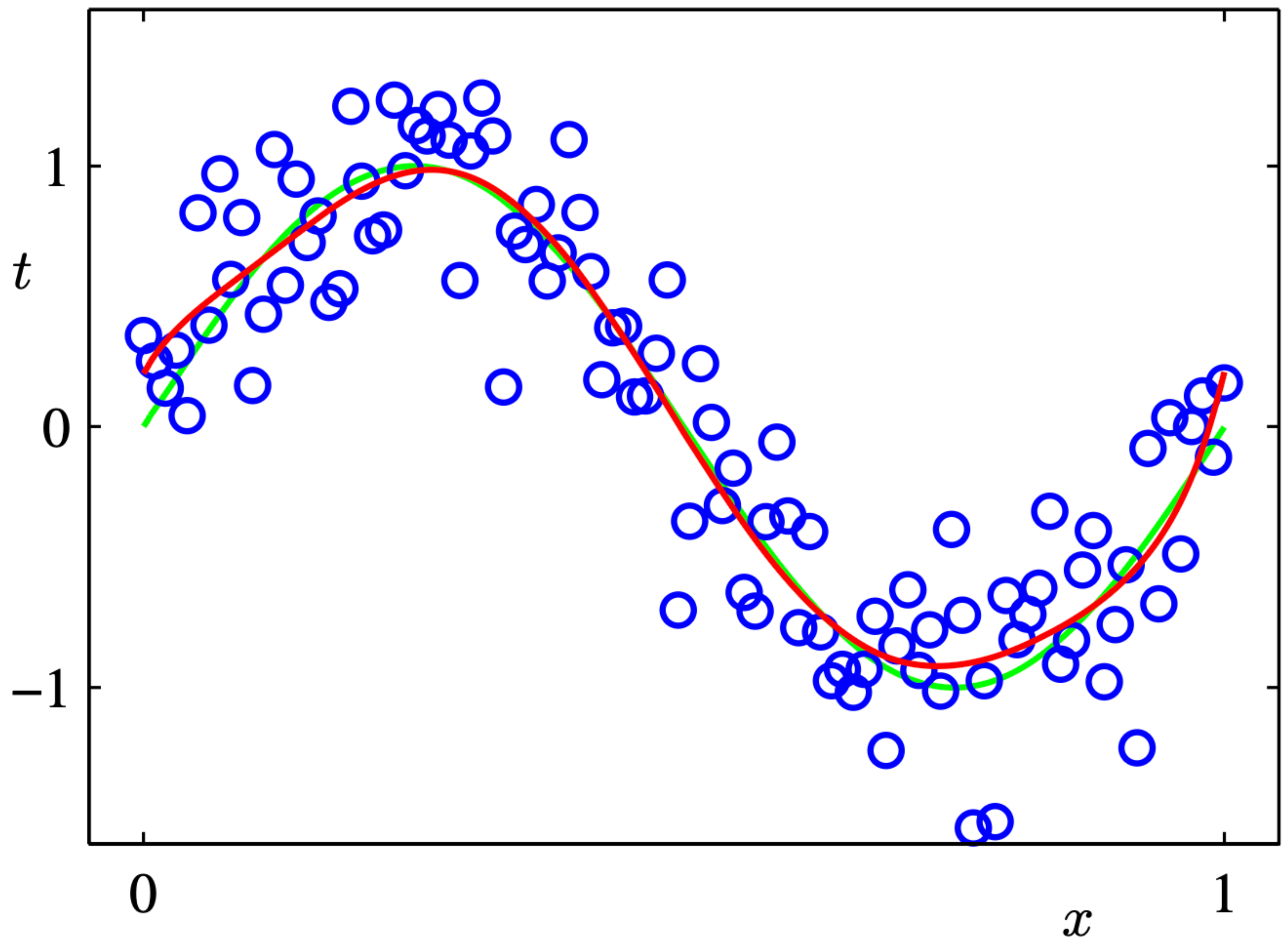
$$p(\mathbf{x}|\mu,\sigma^2) = \prod_{n=1}^{N}\mathcal{N}\left(x_n|\mu,\sigma^2\right)$$

What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty

I.I.D ?



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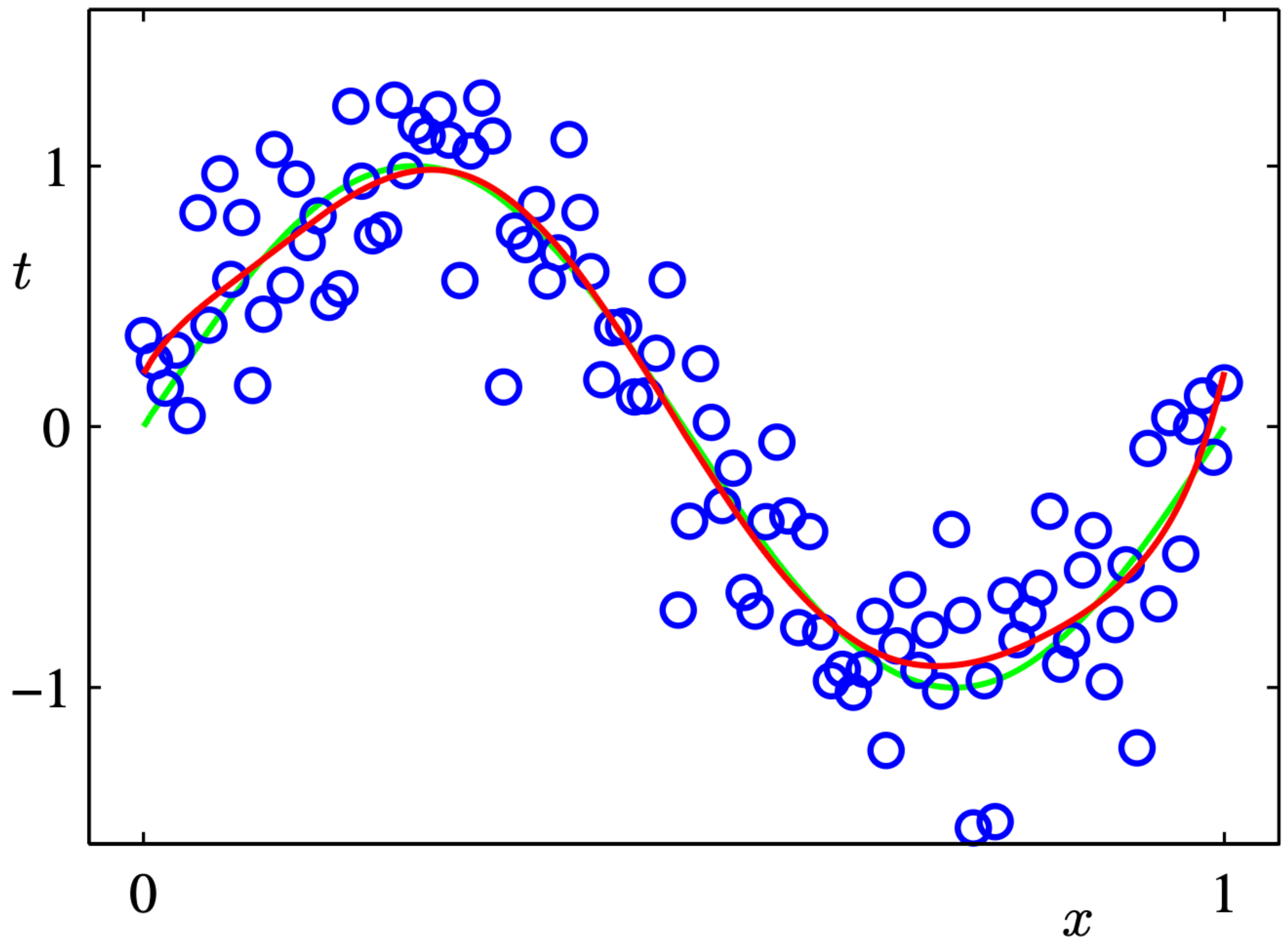
What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty

Prediction (Inference)

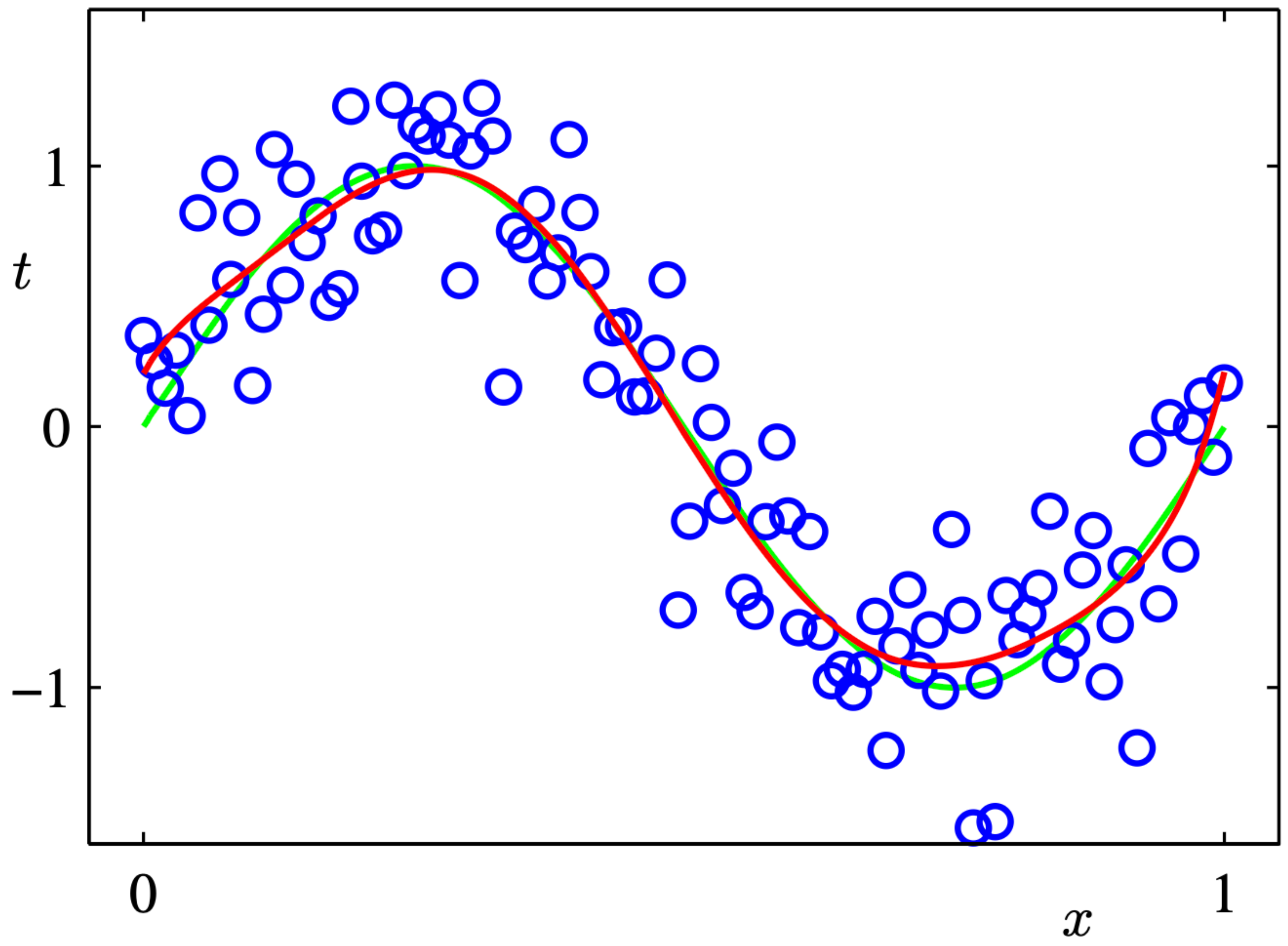


What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty **Prediction (Inference)**

$$\hat{Y} = \hat{f}(X)$$



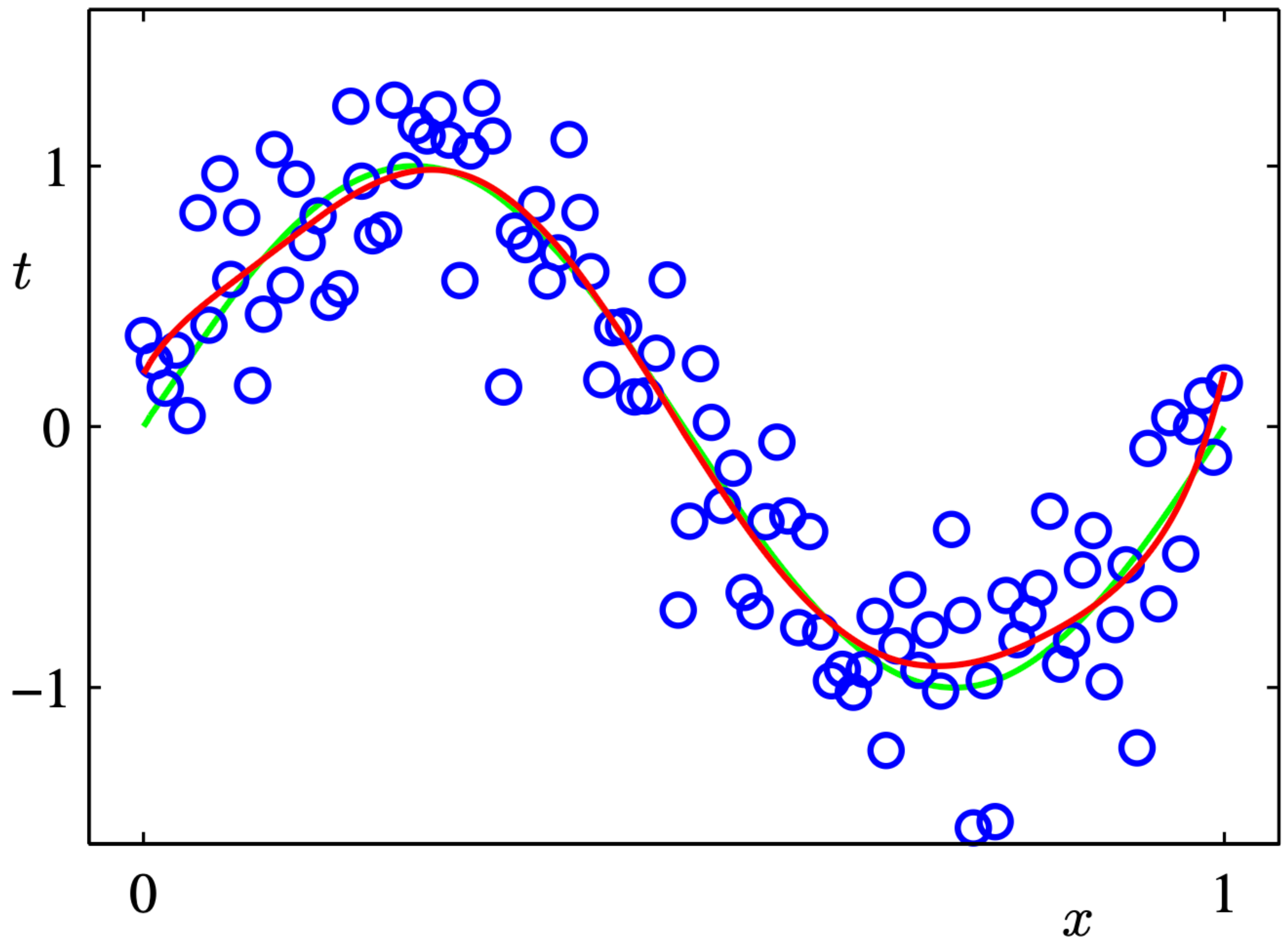
$$Y = f(X) + \epsilon$$

AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty Prediction (Inference) $\hat{Y} = \hat{f}(X)$



$$Y = f(X) + \epsilon$$

$$\begin{aligned} E(Y - \hat{Y})^2 &= E[f(X) + \epsilon - \hat{f}(X)]^2 \\ &= \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{Reducible}} + \underbrace{\mathrm{Var}(\epsilon)}_{\text{Irreducible}} \end{aligned}$$

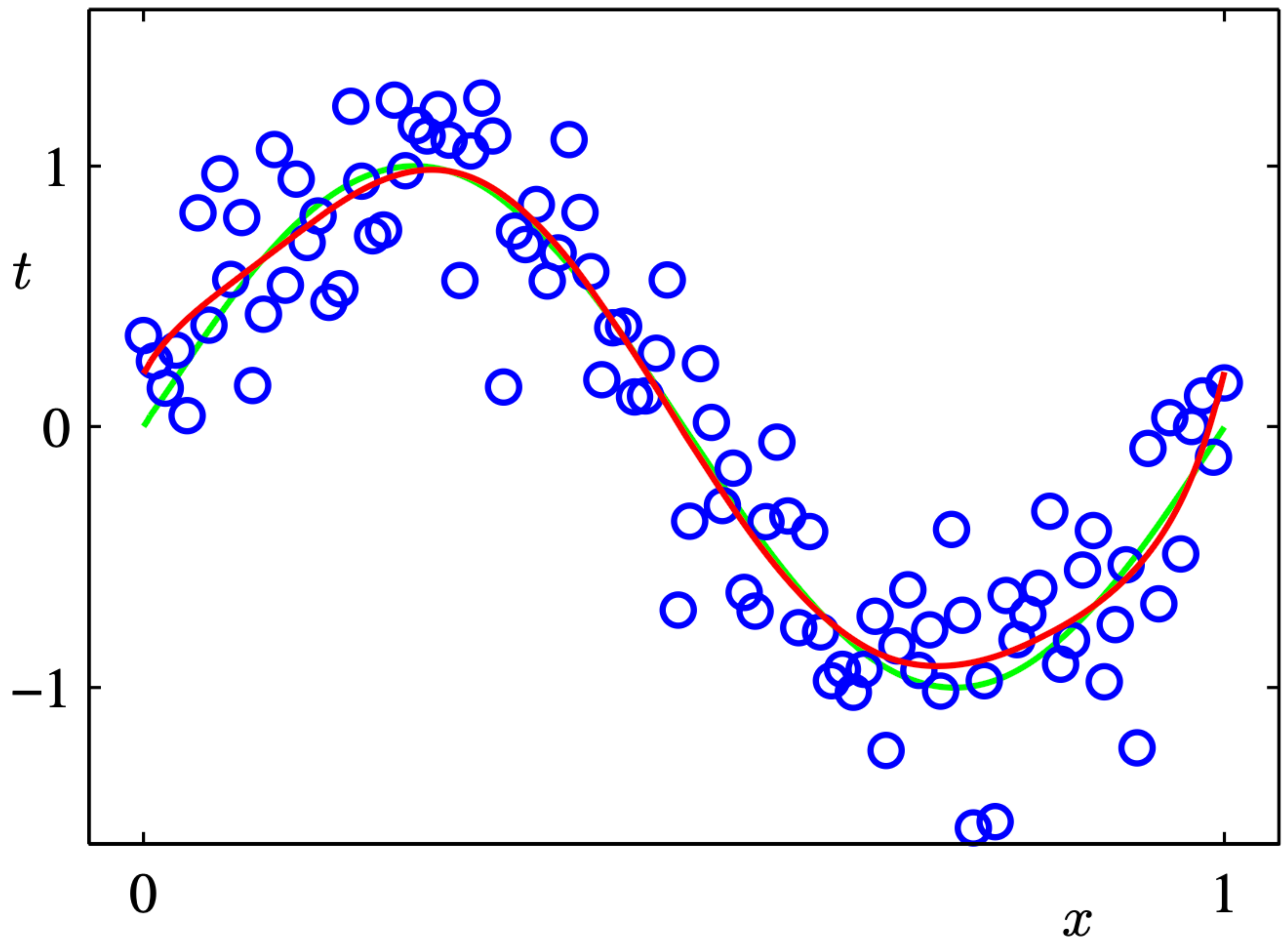
AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Domain Uncertainty **Prediction (Inference)**

$$\hat{Y} = \hat{f}(X)$$



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Show the MLE reduce to squared error loss function

AI-Driven HEP 7

What does an Input look like

An array with shape (n, m, l, k)

Bias - Variance Tradeoff